

When does resource recovery beat a sanitary landfill?

Ex-ante screening of six MSW pathways for 21 Indonesian cities and regencies | per tonne of mixed MSW at the facility gate, 2025 baseline

1 DATA (2025 baseline)

21 cities / regencies

RIPS waste master plans:
composition + tonnage

2025 baseline

Tonnage projected once,
 $Q_{2025} = Q_t (1+g)^{(2025-t)}$

Provenance

Every value labelled; 23 secondary
values re-verified against sources

Parameters

IPCC 2006/2019 defaults, Indonesian
costs, grid factors (ESDM 2018)

2 MODEL

Six options manage the whole tonne (one LCA/TEA boundary)



Climate (GWP100) · net cost · fossil energy · land

Feasibility gates

LHV, scale, RDF quality, kiln distance, PHB scale

Decision

carbon-inclusive cost $C + p \cdot G / 1000$,
fixed $p = 0, 2, 25, 50, 100$ USD/t CO₂e

Uncertainty

Monte Carlo 4,000 draws × 21 locations,
Sobol indices, stress tests

Decision support

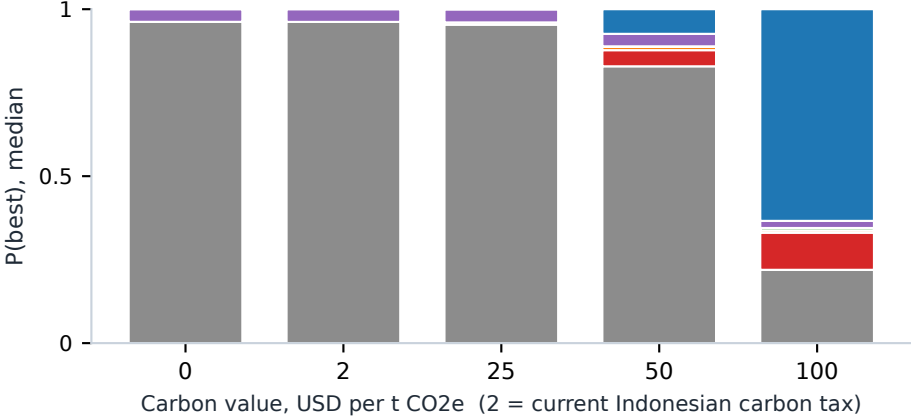
scenario discovery (CART/PRIM), break-even
targets, value of information

Future

2045 projection: grid to net zero in 2060,
real cost escalation

3 RESULTS

Which option is preferred? (21 locations)



Up to 50 USD/t CO₂e

Landfill + flare is cheapest in all 21 locations

At 100 USD/t CO₂e

RDF + AD wins in 20 of 21 (abatement ~73 USD/t CO₂e)

Measure first

waste moisture/composition & landfill-gas capture (EVPI 3.2 USD/t)

In 2045

cleaner grid: WtE GHG +177 kg CO₂e/t; RDF + AD unchanged

Answer to the research question: conditions under which each pathway becomes preferable

Landfill + flare

carbon value ≤ ~50 USD/t,
or no cement kiln in reach

RDF + AD

carbon value near 73-100 USD/t,
kiln ≤ 300 km, waste not too wet

Waste-to-energy

Perpres 109/2025 tariff +
≥ 1,000 t/day + LHV ≥ 7 MJ/kg

Anaerobic digestion

food separated at source +
carbon value > ~30-45 USD/t